# GYANMANJARI INNOVATIVE UNIVERSITY (GMIU)

Department of Computer Engineering & IT | Sem 5 (Session 2026-27)

| | |
|---|---|
| **NAME** | **DIVYARAJSINH CHAVDA** |
| **CLASS** | **Y** |
| **ENROLLMENT NO.** | **240905040018** |
| **SUBJECT** | **SOFTWARE ENGINEERING (BETCE15322)** |
| **ACTIVITY** | **ACTIVE LEARNING ACTIVITY - 1 (ALA-1)** |
| **TOPIC** | **ONLINE LIBRARY MANAGEMENT SYSTEM** |

## 1. Executive Summary & System Requirements Analysis

**Problem Scenario:** The **Online Library Management System** automates book searching, reservations, issuing, returns, overdue fine calculation, and catalog management connecting Members (Students/Faculty), Librarians, and System Administrators.

| Req ID | Category | Requirement Description | Primary Stakeholder |
|---|---|---|---|
| FR-01 | Functional | Member authentication and membership profile validation. | Member / Admin |
| FR-02 | Functional | Search book catalog by title, author, category, or ISBN. | Member |
| FR-03 | Functional | Reserve books online when available or currently borrowed. | Member |
| FR-04 | Functional | Issue book to member with automated due date assignment. | Librarian |
| FR-05 | Functional | Process book return and calculate overdue fines automatically. | Librarian / System |
| FR-06 | Functional | Manage book inventory (Add, Update, Remove book stock). | Librarian |
| NFR-01 | Non-Functional | Performance: Catalog search results render within 1.5 seconds. | System |
| NFR-02 | Non-Functional | Security: Role-Based Access Control (RBAC) enforcing admin rights. | System |

## 2. UML Use Case Diagram (Microsoft Visio 365)

The Use Case diagram below was constructed in **Microsoft Visio 365 Professional** using standard UML 2.5 Use Case stencils. It illustrates system boundaries, actors (Member, Librarian, Admin), and relationship dependencies (<<include>>, <<extend>>).

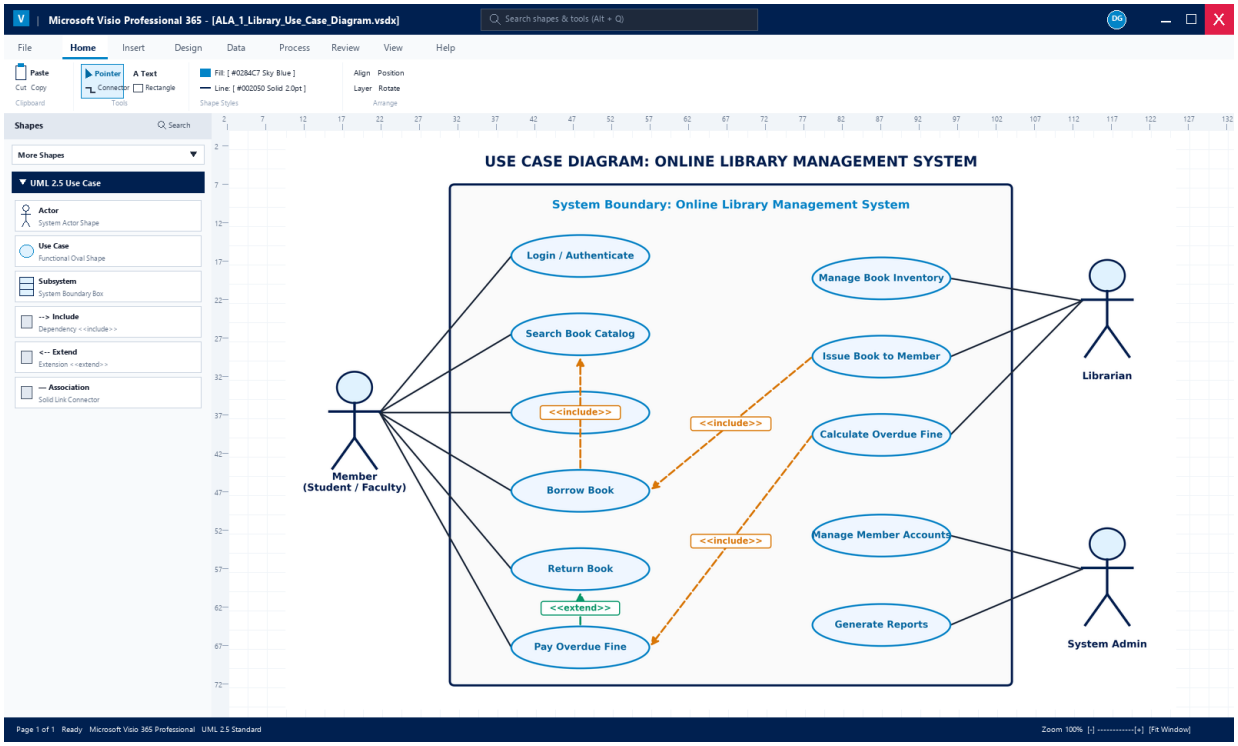



Figure 1: Microsoft Visio 365 Screenshot - Use Case Diagram for Online Library System

| Use Case ID | Use Case Name | Primary Actor | Description & Dependency Flow |
|---|---|---|---|
| UC-01 | Login / Authenticate | Member / Admin | Validates user credentials and grants portal access. |
| UC-02 | Search Book Catalog | Member | Searches inventory; <<include>> dependency in Borrow Book. |
| UC-03 | Reserve Book | Member | Places advance reservation on unavailable or checked-out books. |
| UC-04 | Borrow Book | Member | Initiates book loan request; <<include>> search catalog validation. |
| UC-05 | Return Book | Member | Returns borrowed item; <<extend>> trigger for Pay Fine if overdue. |
| UC-06 | Pay Overdue Fine | Member | Processes fine payment; <<include>> in Calculate Fine process. |
| UC-07 | Manage Inventory | Librarian | Adds, updates, or deletes book records from library database. |

## 3. UML Class Diagram (Microsoft Visio 365)

The Class Diagram below was modeled in **Microsoft Visio 365 Professional** using 3-compartment UML Class stencils. It defines object entities, attributes, methods, generalization inheritance, and multiplicities.

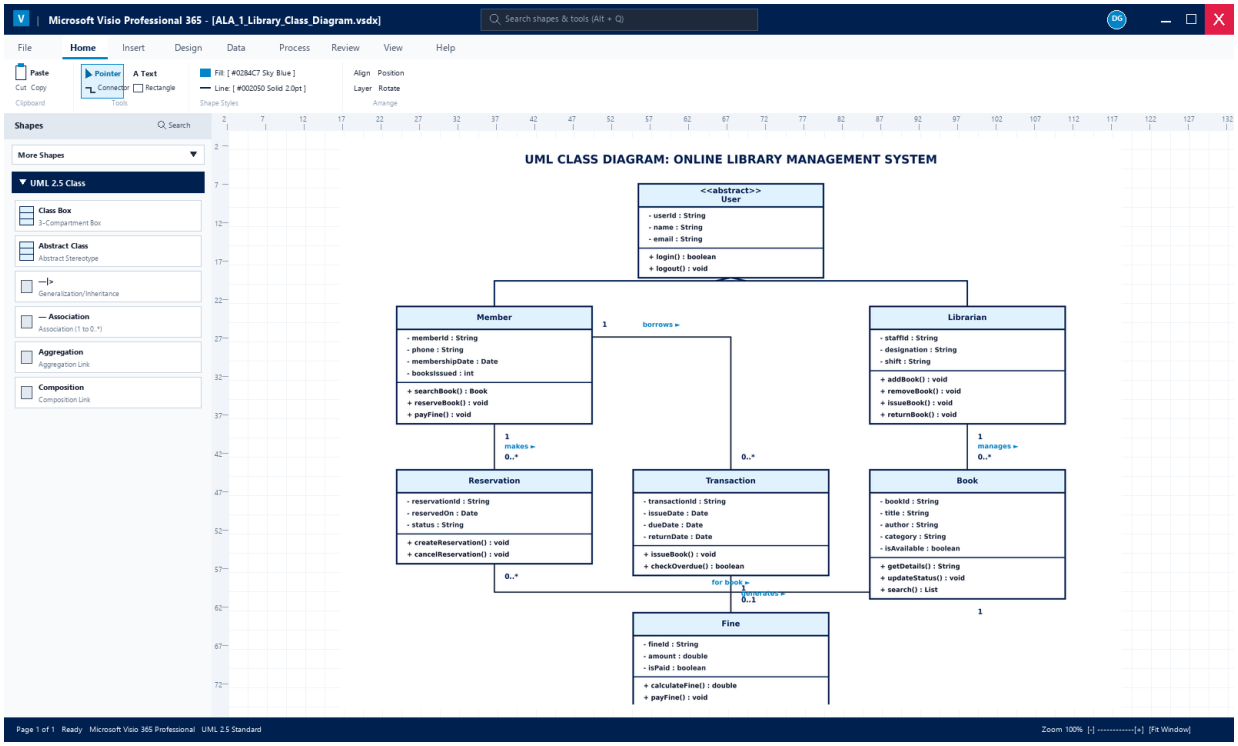



Figure 2: Microsoft Visio 365 Screenshot - Class Diagram showing Domain Entities & Multiplicities

| Class Name | Stereotype / Type | Attributes | Methods | Associations & Multiplicity |
|---|---|---|---|---|
| User | Abstract Superclass | userId, name, email | login(), logout() | Superclass for Member & Librarian |
| Member | Subclass (Inherits User) | memberId, phone, membershipDate | searchBook(), reserveBook(), payFine() | 1 -- 0..* Reservation; 1 -- 0..* Transaction |
| Librarian | Subclass (Inherits User) | staffId, designation, shift | addBook(), issueBook(), returnBook() | 1 -- 0..* Book ('manages') |
| Reservation | Domain Entity | reservationId, reservedOn, status | createReservation(), cancelReservation() | 0..* -- 1 Book ('for book') |
| Transaction | Transaction Entity | transactionId, issueDate, dueDate | issueBook(), checkOverdue() | 1 -- 0..1 Fine ('generates') |
| Book | Domain Entity | bookId, title, author, isAvailable | getDetails(), updateStatus() | Associated with Reservation & Transaction |
| Fine | Financial Entity | fineId, amount, isPaid | calculateFine(), payFine() | 0..1 generated per overdue Transaction |

## 4. UML Sequence Diagram (Microsoft Visio 365)

The Sequence Diagram below was created in **Microsoft Visio 365 Professional** to trace the dynamic time-ordered execution flow for book searching, issuing, returning, and overdue fine calculation.

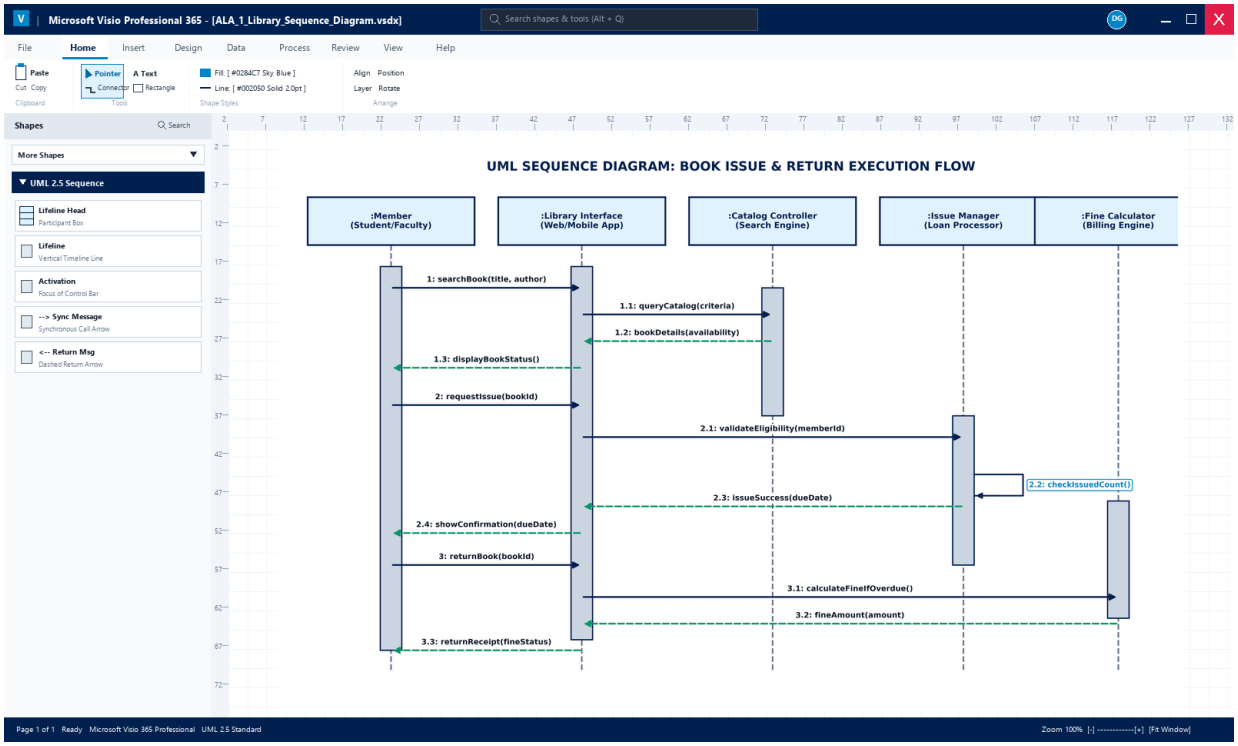



Figure 3: Microsoft Visio 365 Screenshot - Sequence Diagram for Book Issue & Return Execution Flow

| Step # | Source Lifeline | Destination Lifeline | Message Signature / Operation | Message Type |
|---|---|---|---|---|
| 1 | Member | Library Interface | searchBook(title, author) | Synchronous Call |
| 2 | Library Interface | Catalog Controller | queryCatalog(criteria) | Synchronous Call |
| 3 | Catalog Controller | Library Interface | bookDetails(availability) | Return Message |
| 4 | Library Interface | Member | displayBookStatus() | Return Message |
| 5 | Member | Library Interface | requestIssue(bookId) | Synchronous Call |
| 6 | Library Interface | Issue Manager | validateEligibility(memberId) | Synchronous Call |
| 7 | Issue Manager | Issue Manager | checkIssuedCount() | Self Call |
| 8 | Issue Manager | Library Interface | issueSuccess(dueDate) | Return Message |
| 9 | Library Interface | Member | showConfirmation(dueDate) | Return Message |
| 10 | Member | Library Interface | returnBook(bookId) | Synchronous Call |
| 11 | Library Interface | Fine Calculator | calculateFineIfOverdue() | Synchronous Call |
| 12 | Fine Calculator | Library Interface | fineAmount(amount) | Return Message |
| 13 | Library Interface | Member | returnReceipt(fineStatus) | Return Message |

## 5. Conclusion & Tool Verification

All UML diagrams for Active Learning Activity – 1 (ALA-1) were designed using **Microsoft Visio 365 Professional** using standard UML 2.5 shape stencils. The system requirements for the Online Library Management System have been completely modeled into Use Case, Class, and Sequence blueprints as required by the GMIU Software Engineering curriculum.

> 📌 **Student Self-Declaration**
> I, DIVYARAJSINH CHAVDA (Enrollment No: 240905040018, Class Y), hereby confirm that this Active Learning Activity (ALA-1) assignment has been individually authored and modeled in Microsoft Visio 365 according to the guidelines set by the Department of Computer Engineering & IT, Gyanmanjari Innovative University (GMIU).